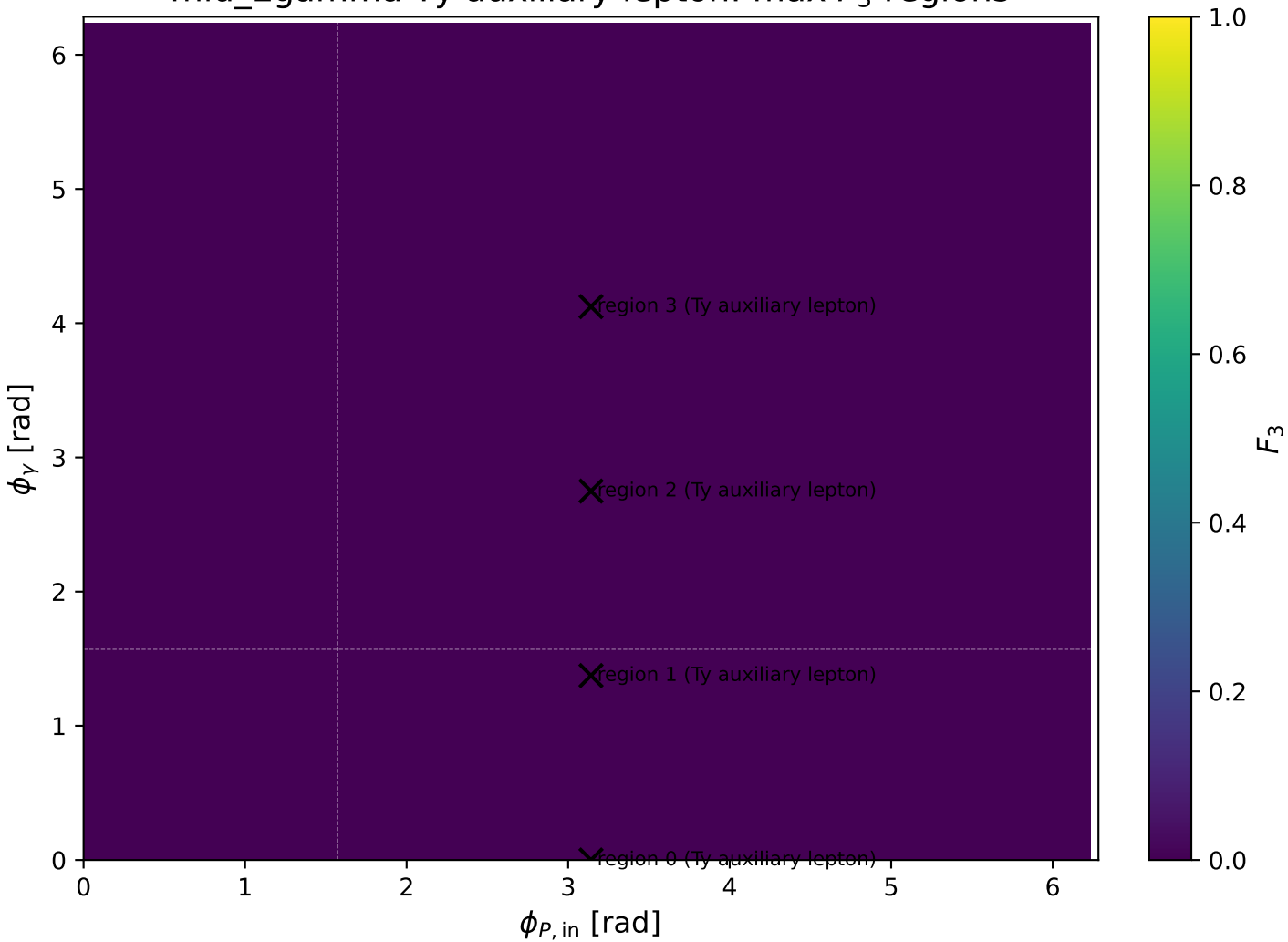
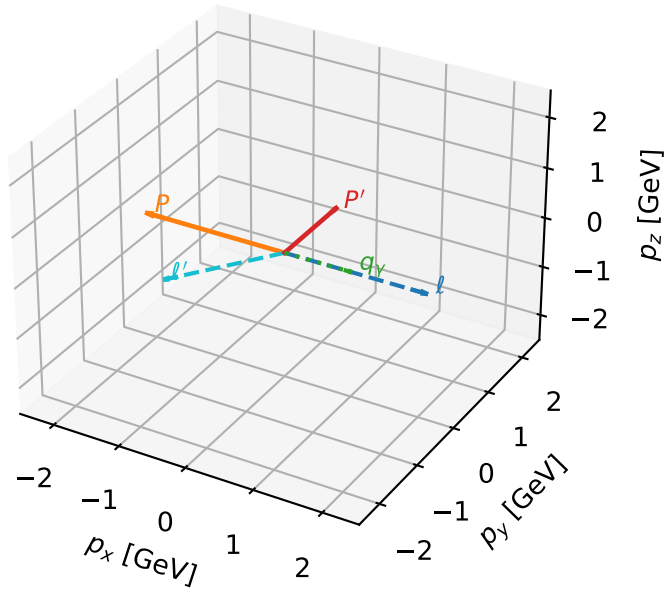


mid_Egamma Ty auxiliary lepton: max F_3 regions



Ty auxiliary lepton characteristic max F_3 configuration

3D momenta

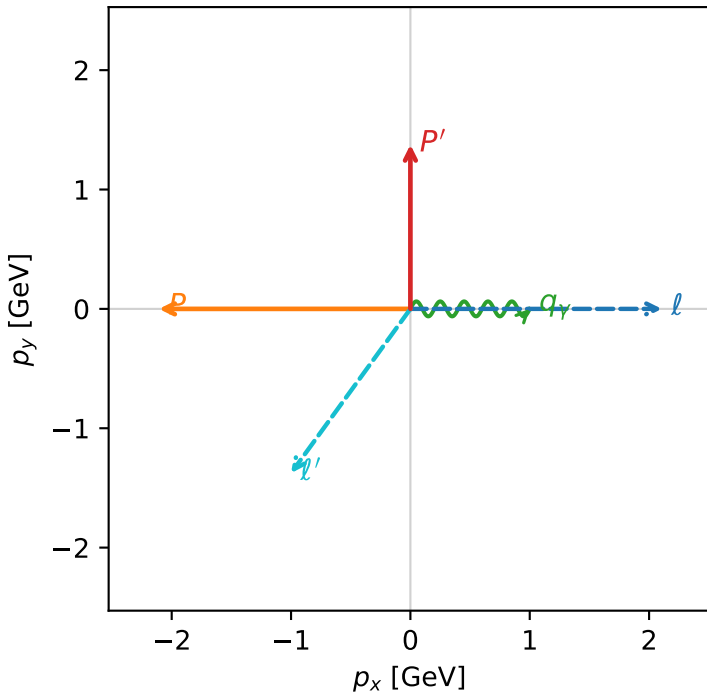


f3_mid_egamma_ty_lepton_region_0 F_3=0
 region: mid_Egamma_Ty_lepton_region_0
 outgoing-state purity: 0.551947
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2} \sum_{h_p} \rho(T_{y,l}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.37604$
 $\theta_{in}=1.57079$, $\phi_l=0$, $\phi_P=3.14159$
 $E_V=1$, $\phi_Y=0$
 $Q^2=11.4177$, $x_B=0.774167$, $t=-5.92183$, $W^2=4.2105$, $y=0.751086$
 $\theta(l', \gamma)=126.007$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 2.1069$ $p_y= -0.0000$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -2.1069$ $p_y= 0.0000$ $p_z= 0.0000$
 l' $E= 1.9732$ $p_x= -1.0000$ $p_y= -1.3760$ $p_z= 0.0000$
 P' $E= 1.6653$ $p_x= 0.0000$ $p_y= 1.3760$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 1.0000$ $p_y= 0.0000$ $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=-1, h_Y=+1$
 electron Ty, proton h=-1

$h_e=+1, h_p=+1, h_Y=-1$
 electron Ty, proton h=+1

$h_e=-1, h_p=+1, h_Y=+1$
 electron Ty, proton h=-1

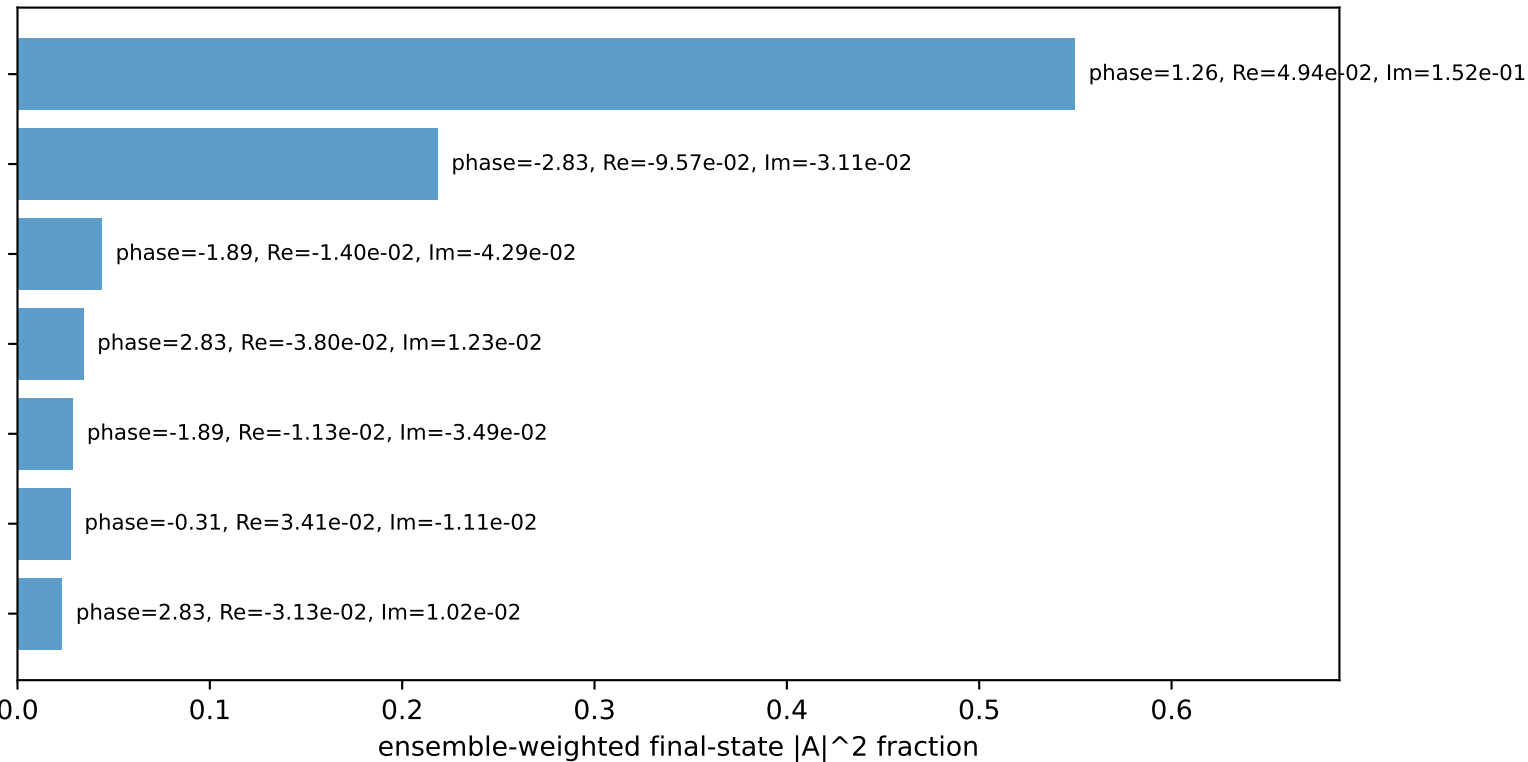
$h_e=-1, h_p=+1, h_Y=+1$
 electron Ty, proton h=+1

$h_e=-1, h_p=-1, h_Y=-1$
 electron Ty, proton h=-1

$h_e=-1, h_p=+1, h_Y=-1$
 electron Ty, proton h=+1

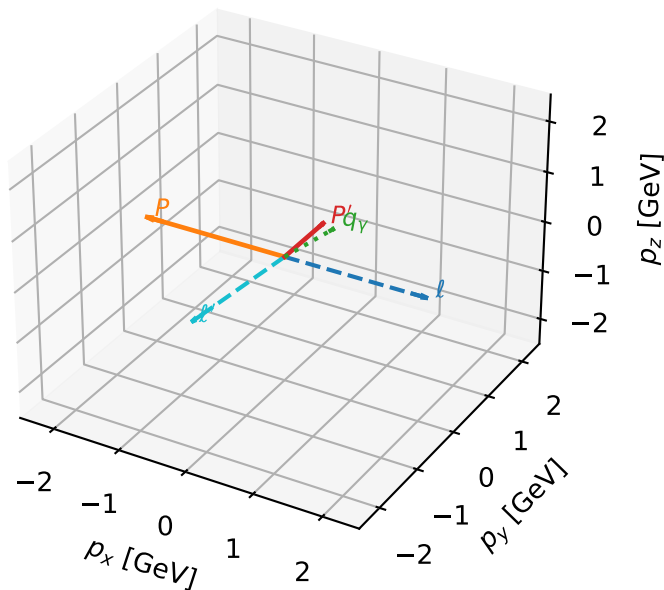
$h_e=-1, h_p=-1, h_Y=+1$
 electron Ty, proton h=+1

Leading initial-ensemble/final-state components (N=7, retained=92.7%)



Ty auxiliary lepton characteristic max F_3 configuration

3D momenta

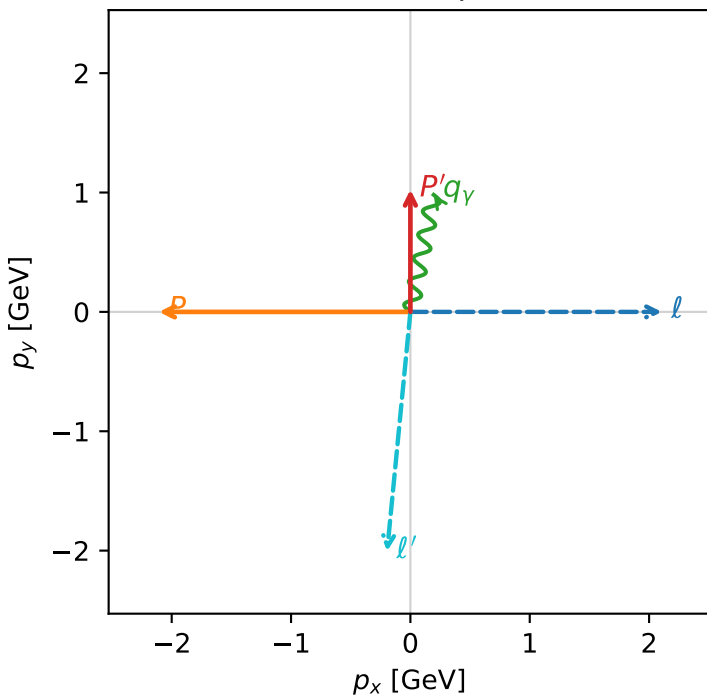


f3_mid_egamma_ty_lepton_region_1 F_3=0
 region: mid_Egamma_Ty_lepton_region_1
 outgoing-state purity: 0.501629
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2} \sum_{h_p} \rho(T_{y,l}, h_p)$

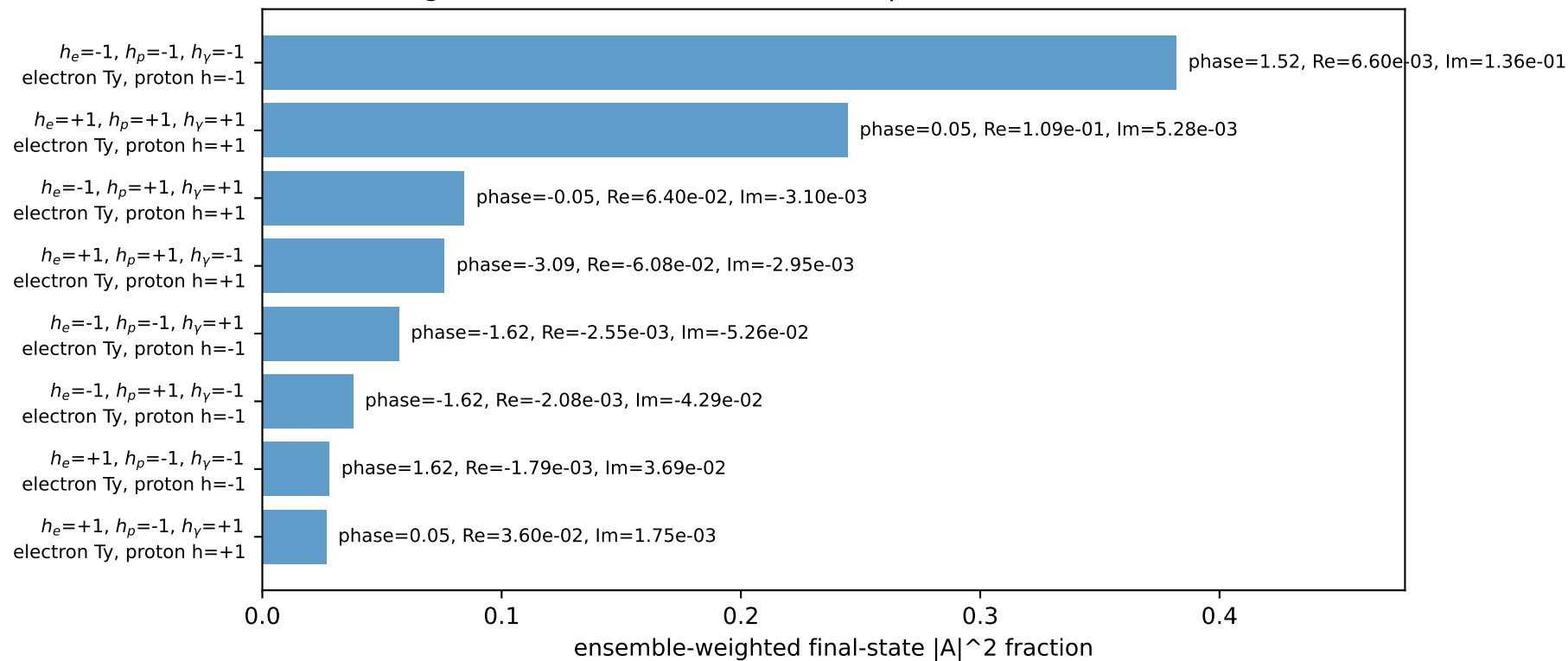
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.02462$
 $\theta_{in}=1.57079$, $\phi_l=0$, $\phi_p=3.14159$
 $E_V=1$, $\phi_V=1.37445$
 $Q^2=9.31416$, $x_B=0.923789$, $t=-4.64785$, $W^2=1.64824$, $y=0.513473$
 $\theta(l', \gamma)=174.306$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 2.1069$ $p_y= -0.0000$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -2.1069$ $p_y= 0.0000$ $p_z= 0.0000$
 l' $E= 2.2494$ $p_x= -0.1951$ $p_y= -2.0054$ $p_z= 0.0000$
 P' $E= 1.3891$ $p_x= 0.0000$ $p_y= 1.0246$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.1951$ $p_y= 0.9808$ $p_z= 0.0000$

Transverse plane

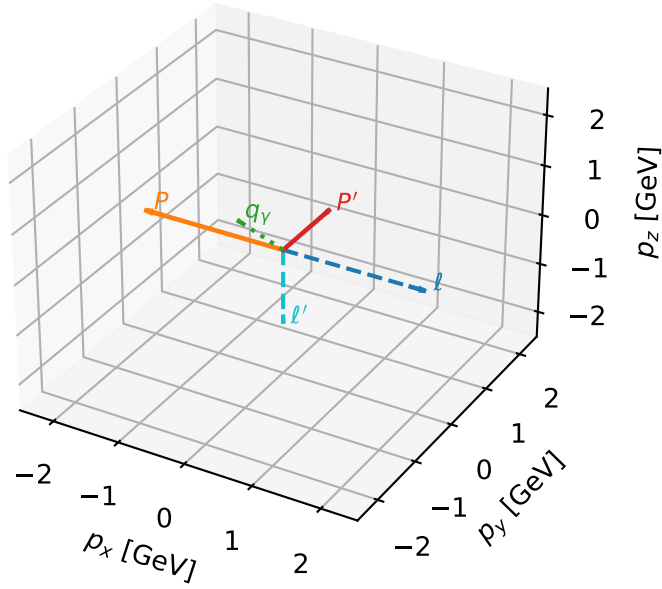


Leading initial-ensemble/final-state components (N=8, retained=93.7%)



Ty auxiliary lepton characteristic max F_3 configuration

3D momenta

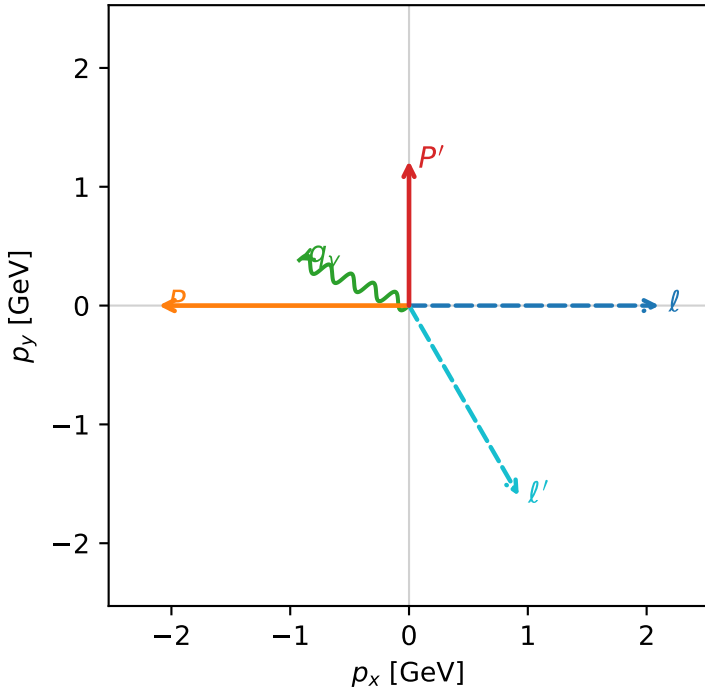


f3_mid_egamma_ty_lepton_region_2 F_3=0
 region: mid_Egamma_Ty_lepton_region_2
 outgoing-state purity: 0.501646
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,l},h_p)$

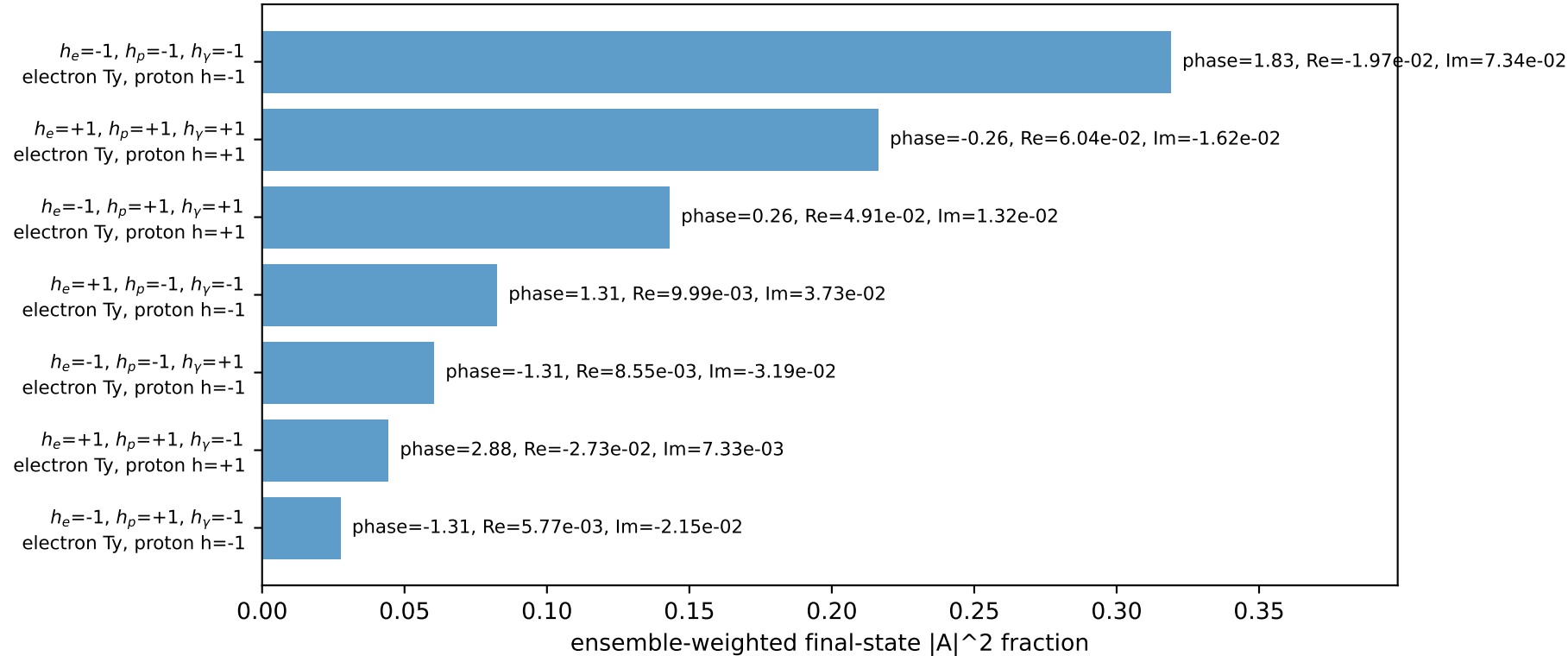
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.21789$
 $\theta_{in}=1.57079$, $\phi_l=0$, $\phi_p=3.14159$
 $E_V=1$, $\phi_V=2.74889$
 $Q^2=3.90814$, $x_B=0.645922$, $t=-5.33099$, $W^2=3.02219$, $y=0.308132$
 $\theta(l',\gamma)=142.494$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 2.1069$ $p_y= -0.0000$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -2.1069$ $p_y= 0.0000$ $p_z= 0.0000$
 l' $E= 2.1013$ $p_x= 0.9239$ $p_y= -1.6006$ $p_z= 0.0000$
 P' $E= 1.5372$ $p_x= 0.0000$ $p_y= 1.2179$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.9239$ $p_y= 0.3827$ $p_z= 0.0000$

Transverse plane

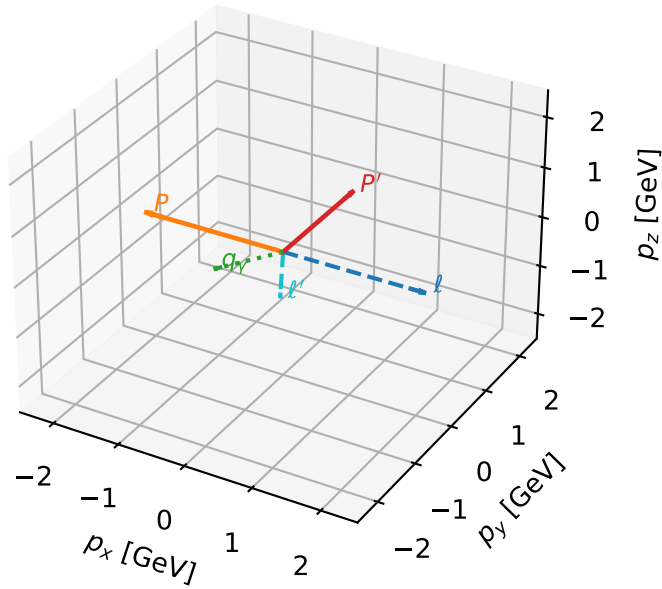


Leading initial-ensemble/final-state components (N=7, retained=89.3%)



Ty auxiliary lepton characteristic max F_3 configuration

3D momenta



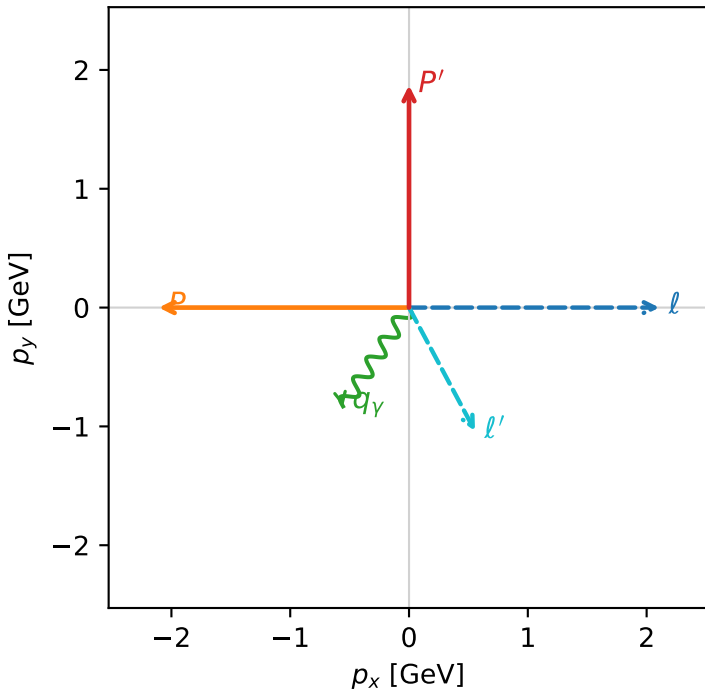
f3_mid_egamma_ty_lepton_region_3 F_3=0
 region: mid_Egamma_Ty_lepton_region_3
 outgoing-state purity: 0.500574
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,l},h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.87089$
 $\theta_{in}=1.57079$, $\phi_l=0$, $\phi_p=3.14159$
 $E_V=1$, $\phi_V=4.12334$
 $Q^2=2.86848$, $x_B=0.282181$, $t=-7.89387$, $W^2=8.17674$, $y=0.517691$
 $\theta(l',\gamma)=61.8745$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.3322$ $p_x= 2.1069$ $p_y= -0.0000$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -2.1069$ $p_y= 0.0000$ $p_z= 0.0000$
 l' $E= 1.5457$ $p_x= 0.5556$ $p_y= -1.0394$ $p_z= 0.0000$
 P' $E= 2.0929$ $p_x= 0.0000$ $p_y= 1.8709$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.5556$ $p_y= -0.8315$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=8, retained=91.2%)

